

2.1.1 Physical robot

(10 points) The final product must be designed as a chain RRR manipulator, with three active joints. It must be equipped with a marker or a pen, which will draw on the paper. The pen/marker/pencil can be permanently attached to the robot with various adhesives. The designers are encouraged to consider different RRR configurations and pay special care to the orientation of the pen/marker/pencil. The robot must be designed using the provided mechanical components and the 3D printed parts. Please consult with your lecturers before using any additional equipment and/or materials.

2.1.2 Machining components

(10 points) Your components will be machined for you, by a 3D printer. Upon submission of your designs, only feasibility of manufacturing will be checked. If design is acceptable for 3D printing, it will be printed. Students are ultimately responsible for the design (dimensions, construction, etc...). Your design will be compared with other groups based on the amount of material used to print objects. The amount of material is measured according to how many filament meters have been used to print your parts and will be a factor in scoring the points for this category.

2.1.3 Camera

(5 points) The camera frame and the robot base frame must be properly calibrated. You may use passive markers, both mechanical or algorithmic, to calibrate the camera with respect to the robot base coordinate frame.

2.1.4 Direct kinematics

(5 points) You must derive a direct kinematics algorithm which calculates the position of the end effector from input joint values. You must take into account the size, shape and the structure of the robot. You must be able to localize the robot end-effector precisely within its workspace.

2.1.5 Inverse kinematics

(5 points) You must derive an inverse kinematics algorithm which controls the joints given the position of the end effector of the robot. You must take into account the size, shape and the construction of the robot. You must be able to position a robot end-effector precisely within its workspace.

2.2 User Interface

Your final product must include a practical and effective user interface, so that an unskilled operator (aka a professor and/or assistant) can operate the robot.

2.2.1 GUI design

(5 points) The graphical user interface must include the standard GUI elements, such as buttons, sliders, etc., in an intuitive, logical and easy-to-use design. It must enable switching from manual to autonomous mode. The autonomous mode must have a text box for natural language input.

2.2.2 Manual control - direct kinematics

(5 points) In manual control, the GUI must be designed so that the operator can directly enter the joint values, both via textbox and via slider. The robot must move to the designated values, and the GUI must output the 3D position of the end effector for the current joint values.

2.2.3 Manual control - inverse kinematics

(5 points) In manual control mode, GUI must be designed so that the operator can enter the value for the position of the end effector. The robot must move to the designated position, and the GUI must output the value of all joints at all times.

2.2.4 Robot URDF model

(5 points) The URDF model of your robot must be visualized in RVIZ, showing the current configuration of the robot. There is no requirement to visualize the current configuration of the robot in your GUI, but that does not mean you are not allowed to do it.

2.2.5 Autonomous mode - natural language

(5 points) When in autonomous mode, your interface must be able to process a natural language command entered in a text box.

2.3 Autonomous drawing

2.3.1 Robot vision

(5 points) Your robotic system must be capable of recognizing and localizing objects, which can attain different positions and orientations within the workspace of the robot.

2.3.2 Tool path planning

(5 points) Your system must be able to plan paths for the end effector. Path planning can occur in any suitable space (image space, joint space, task space) but needs to be converted into the task space (path of the end-effector) and executed using inverse kinematics.

2.3.3 Autonomous mode

(10 points) Once the system receives the natural language instruction consisting of objects to be connected and objects to be avoided, the system must be able to recognize objects that need to be connected and the objects that need to be avoided, and plan and execute the path of the marker tip accordingly.

2.4 Time challenge

(20 points) In the final test, you will be provided with a language command that includes objects to be connected and objects to be avoided. You will have a limited amount of time to complete this task. The total time to completion will be measured. The best ranking team will receive 20 points, the second best team will receive a bit less than 20 points, and so on. The quality of the

line and its length will be assessed. Crossing the objects that must be avoided will result in a penalty. In case of a tie, both teams will receive the points that would be awarded to the higher rank. You can complete some parts of the task manually if needed; however, every autonomous result is ranked higher than all attempts that are deemed manual.

Points

There is a total of 100 points in the FAT. When calculating the total cumulative for the final grade, the points will be scaled as announced at the beginning of lectures.

Example

A successful example from last year (group SEM2) is provided in Figure 2.



Figure 2: Successful execution of the task "connect stop sign and the car, avoid cats and traffic lights".